21 October 2026

The Editors
Sequential Analysis

Dear Editors,

Please consider the enclosed manuscript, "Random-walk level noise breaks anytime-valid trend monitoring: A failure theorem, a universal e-process and the price of robustness", as an original article for Sequential Analysis.

Anytime-valid monitoring of a trend assumes independent noise around a smooth path. The manuscript proves that this fails under the smallest departure from independence: if the latent level carries a random-walk disturbance of any positive variance, every fixed-shape contrast statistic has its variance inflated in proportion to the square of the series length, and e-processes built on independent noise declare a change with probability tending to one. It then gives a universal e-process, valid for any consistent parametric Gaussian covariance class with unknown scale and linear trend, built on the maximal invariant under location and scale, with a proof of validity. A third result is an information ceiling that bounds, for every valid monitor, the expected log e-value and the power; the e-process falls short of the ceiling by about the logarithm of its mixture size, which we measure. The method is applied to the 56 published waves of the U.S. Census Bureau's Business Trends and Outlook Survey on business use of artificial intelligence, where the independence-based monitor declares changes in nearly every series simulated from the fitted null and the universal e-process reaches an e-value of 18 against a threshold of 20. We report this as a near miss.

The manuscript is original, is not under consideration elsewhere and has not been published. It is anonymized, with a separate title page. The data are public and the code reproduces every number, table and figure. Generative AI tools were used to support coding and to polish the language; the ideas, methods and algorithms are mine, and I reviewed and edited all output and take full responsibility for the work. I have no competing interests to declare and I have not suggested reviewers.

Yours sincerely,

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